

Thermodynamic characterization of the marine boundary layer during fog and dew events in the hyperarid coastal climate of the Atacama Desert

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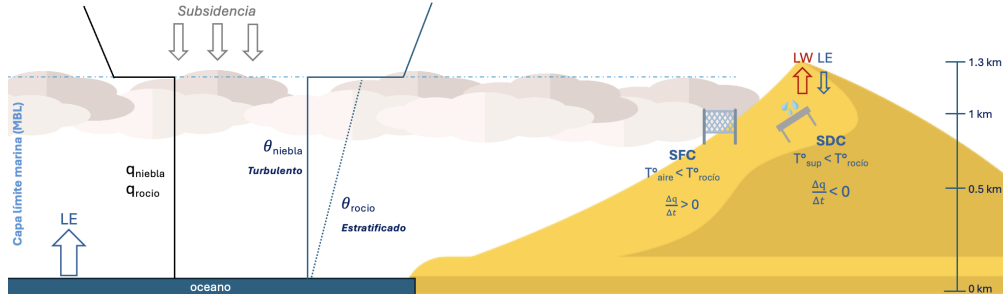
August 10, 2026

1 Introduction

- In the coastal Atacama Desert, precipitation is very low, so fog and dew are the main water sources sustaining the hyperarid ecosystems of *Tillandsia landbeckii* (and other species) and supplementing the water supply of local communities.
- Standard mechanical collectors (SFC for fog, SDC for dew) fail to separate the two phenomena; there is instrumental overlap between collectors, which leaves their relative contribution highly uncertain. A fog collector can generate dew on any of its structures, and a dew collector can capture fog deposition.
- Knowledge of dew is limited, and it is important to study it further and differentiate it from fog, in order to understand the ecosystem's hydrology and also to develop better collection technologies for local communities, enabling the development of better water-related public policies in arid regions.
- There is an alternative for differentiating fog and dew. Some studies exist on regional stability thresholds ($\Delta\theta/\Delta z$), which have been calibrated exclusively to separate “fog” from “non-fog”; dew has not yet been treated as an independent thermodynamic regime.
- Research question: How do atmospheric thermodynamics and surface-atmosphere coupling differ between fog and dew events along the altitudinal transect of the coastal Atacama desert?
- Hypotheses, formulated as falsifiable predictions:
 - *H1 (occurrence)*. The actual percentage of dew events significantly exceeds that indicated by the collector records.
 - *H2 (structure)*. Fog corresponds to a more well-mixed marine boundary layer (lower $\Delta\theta/\Delta z$); dew to a more stratified layer (higher $\Delta\theta/\Delta z$). $\Delta q/\Delta z$ and $\Delta U/\Delta z$ are predicted not to discriminate between regimes, since both phenomena require a nearly saturated, well-mixed vapor column to form.
 - *H3 (dynamics)*. Fog increases specific humidity over time ($\Delta q/\Delta t > 0$, continuous marine advection); dew reduces it or keeps it close to neutral ($\Delta q/\Delta t < 0$, surface moisture sink), reflecting opposite directions of the surface latent heat flux.

2 Physical processes of fog and dew under study

Comentario: La siguiente figura fue modificada y el colector de rocío está más al medio, para que no de a confusión de que está hacia el interior (comentario Camilo tesis), igual quizás puede mejorarse para un paper



- Fog and dew are condensation processes in the atmospheric boundary layer (ABL), governed by turbulent mixing, advection, and thermal stratification.
- To characterize stability independently of compression effects and thermal fluctuations, quasi-conserved variables are used: potential temperature (θ) and specific humidity (q).
- θ (K): the temperature an air parcel would have if brought adiabatically to $P_0 = 1000$ hPa; its vertical gradient ($\Delta\theta/\Delta z$) measures static stability (low = mixed layer, high = stable layer).
- q (g kg^{-1}): mass of vapor per mass of moist air; conserved except for condensation/evaporation, it tracks vapor content and fluxes independently of temperature.
- θ and q diagnose, respectively, vertical thermal stability and moisture sources, allowing fog to be distinguished from dew.
- Fog forms from supersaturated air in contact with the surface ($T_{\text{air}} \leq T_{\text{dew}}$); in northern Chile it is associated with advection of marine stratocumulus, generated by the Humboldt Current (cold SST) plus subsidence from the South Pacific Anticyclone.
- This interaction generates a strong thermal inversion ($\Delta\theta \approx 20$ K) that limits vertical mixing and favors inland transport via the sea breeze, within a band of 500–1200 m a.s.l. and up to 50–60 km from the coast.
- Fog occurs under a well-mixed MBL: $\Delta\theta/\Delta z < 3.1 \times 10^{-3} \text{ K m}^{-1}$; $\Delta q/\Delta t \approx +0.016 \text{ g kg}^{-1} \text{ h}^{-1}$ (continuous moisture advection).
- Dew forms through nocturnal radiative cooling of the surface below the dew point ($T_{\text{surf}} \leq T_{\text{dew}}$); it is direct condensation on a solid surface, not in the air, typical of clear skies, weak winds, and high relative humidity.
- Dew is associated with greater stability. A study with very limited data indicates: $\Delta\theta/\Delta z \approx 3.8 \times 10^{-3} \text{ K m}^{-1}$; $\Delta q/\Delta t \approx -0.25 \text{ g kg}^{-1} \text{ h}^{-1}$ (net vapor transfer toward the surface).
- $\Delta\theta/\Delta z$ thresholds for distinguishing fog from non-fog are highly predictive (91% success); $\Delta q/\Delta z$ is poorly predictive (65%) due to vapor homogeneity within the MBL.

- Previous studies set thresholds of $\Delta\theta/\Delta z \approx 0.001 \text{ K m}^{-1}$ (fog) and 0.0038 K m^{-1} (dew) to segregate both phenomena, but based on very limited datasets.
- Existing robust indicators (horizontal visibility, satellite low clouds) only validate fog vs. non-fog, without discriminating their overlap with dew. This gap justifies the study.

3 Methods

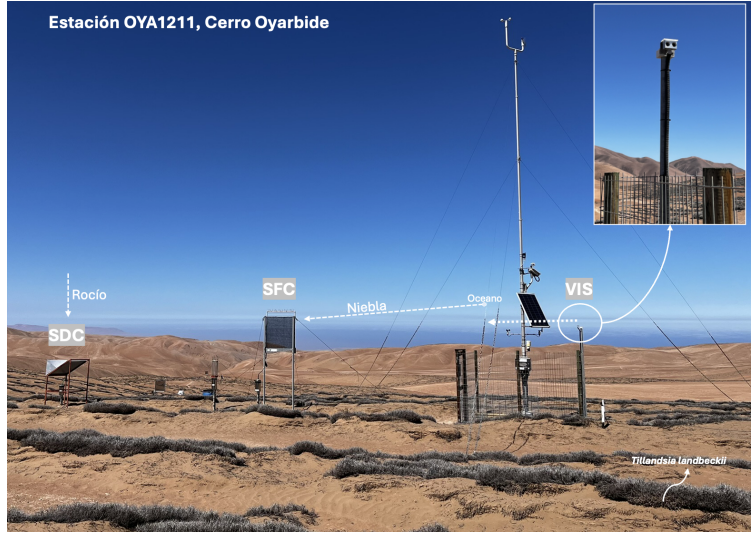
3.1 Study area: Cerro Oyarbide

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- Cerro Oyarbide ($\sim 20.61^\circ \text{ S}$, 70.07° W) is located in the Coastal Cordillera, above Iquique’s Diego Aracena International Airport and $\sim 11 \text{ km}$ inland from the coastline; its hyperarid conditions allow the development of *Tillandsia landbeckii* loma fields between 900 and 1300 m a.s.l.
- It functions as a natural laboratory for this problem: a continuous altitudinal gradient of 8 stations (518–1354 m a.s.l.) crosses exactly the band where regional stratocumulus fog concentrates ($\sim 750 \text{ m a.s.l.}$, Cereceda et al. 2008) and where fog and dew co-occur most frequently — the ambiguity this study seeks to resolve.
- The main OYA1211 station is located within one of these *Tillandsia landbeckii* patches, allowing direct characterization of the atmosphere-biosphere interaction at the same point where the classification protocol is anchored (Fig. 3).
- The coastal baseline is built from the records of Iquique’s Diego Aracena International Airport (48 m a.s.l.), a station latitudinally aligned with the transect.

3.2 Data and instrumentation



- Full-year dataset (2024) at 10-minute resolution, processed in R with tidyverse; variables: air temperature, relative humidity, wind, horizontal visibility, pressure, global radiation, fog/dew collection.
- Data availability varies by station and sensor (Table 1): OYA518, OYA780, OYA1128, and OYA1211 maintain near-complete continuity during 2024, while OYA1069 shows critical gaps due to sensor malfunction (total loss of temperature, 16.7% availability for relative humidity and wind direction), excluding it from the subsequent thermodynamic gradients.
- OYA1211 is the most fully instrumented station along the transect and the only one that brings together, within the same microenvironment, the standard dew collector (SDC), the standard fog collector (SFC), and the horizontal visibility sensor.
- GOES-16 satellite observations are integrated (processed in Google Earth Engine, 2 km spatial resolution, synchronized to 10 min) to detect low clouds (FLC) using a dual-band brightness-temperature threshold algorithm, used as a secondary layer and for regional validation of the classification tree.

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3.3 Hierarchical classification of fog and dew events

- Decision tree anchored at OYA1211: visibility < 1000 m $\rightarrow$ fog; visibility ≥ 1000 m $\rightarrow$ dew; GOES-16 FLC as a secondary criterion when visibility is unavailable, given its coarser spatial resolution and vertical uncertainty (a detected low-cloud pixel does not necessarily intersect the topography).
- Manual day-by-day quality-control review throughout 2024, cross-checking the automatic classification against the visibility curve and the GOES signal, correcting isolated errors and documenting days of atypical behavior.
- The output of this algorithm, applied day by day throughout 2024, is illustrated with representative cases in Fig. 4 (Results), already with the classification resolved.

Table 1: Percentage data availability (%) for 2024 at a 10-minute temporal resolution across the ground monitoring network. The numeric suffix in each station’s name designates its operating altitude (m a.s.l.). Abbreviations: Temp (air temperature), RH (relative humidity), U (wind speed), WD (wind direction), Fog (surface fog event validation), Pres (atmospheric pressure), Rad (global radiation), Vis (visibility), Dew (dew collection). Dashes (–) indicate that the variable is not monitored at that station. GOES-16 satellite imagery maintained 100% availability for low-cloud detection (FLC) throughout the study period.

Station	Temp (°C)	RH (%)	U (m/s)	WD (°)	Fog (mm)	Pres (hPa)	Rad (W/m ²)	Vis (m)	Dew (mm)
AEROP.	92%	92%	15%	0%	–	92%	0%	0%	–
OYA518	99%	99%	97%	99%	99%	99%	99%	0%	–
OYA780	99%	99%	98%	99%	99%	99%	0%	0%	–
OYA862	86%	91%	86%	91%	9%	0%	0%	0%	–
OYA1069	0%	17%	0%	17%	17%	0%	0%	0%	–
OYA1128	100%	100%	100%	100%	100%	0%	0%	0%	–
OYA1193	90%	92%	90%	92%	92%	0%	0%	0%	–
OYA1211	100%	100%	100%	100%	100%	100%	100%	93%	100%
OYA1354	83%	83%	83%	83%	92%	0%	0%	0%	–

3.4 Calculation of thermodynamic gradients and temporal rates

- Derivation of θ and q : hypsometric equation for pressure at missing levels (anchored to the coastal baseline), saturation/actual vapor pressure equation, using the specific heat of dry air $C_p = 1004.67 \text{ J kg}^{-1}\text{K}^{-1}$.
- Vertical gradients $\Delta\varphi/\Delta z$ ($\varphi = \theta, q, U$) calculated along the 48–1354 m transect relative to the coastal baseline — the quantity that will operationalize H2 in Fig. 9.
- Temporal rates $\Delta\varphi/\Delta t$ using forward finite differences: fixed lags of 10 min up to 180 min for the response at event onset (Fig. 10a); 1-hour differences (tolerance up to 2 h) for diurnal-cycle statistics and the annual distribution (Fig. 10b–c) — the quantity that will operationalize H3.
- All timestamps are processed in UTC (mainland Chile local time = UTC–4).

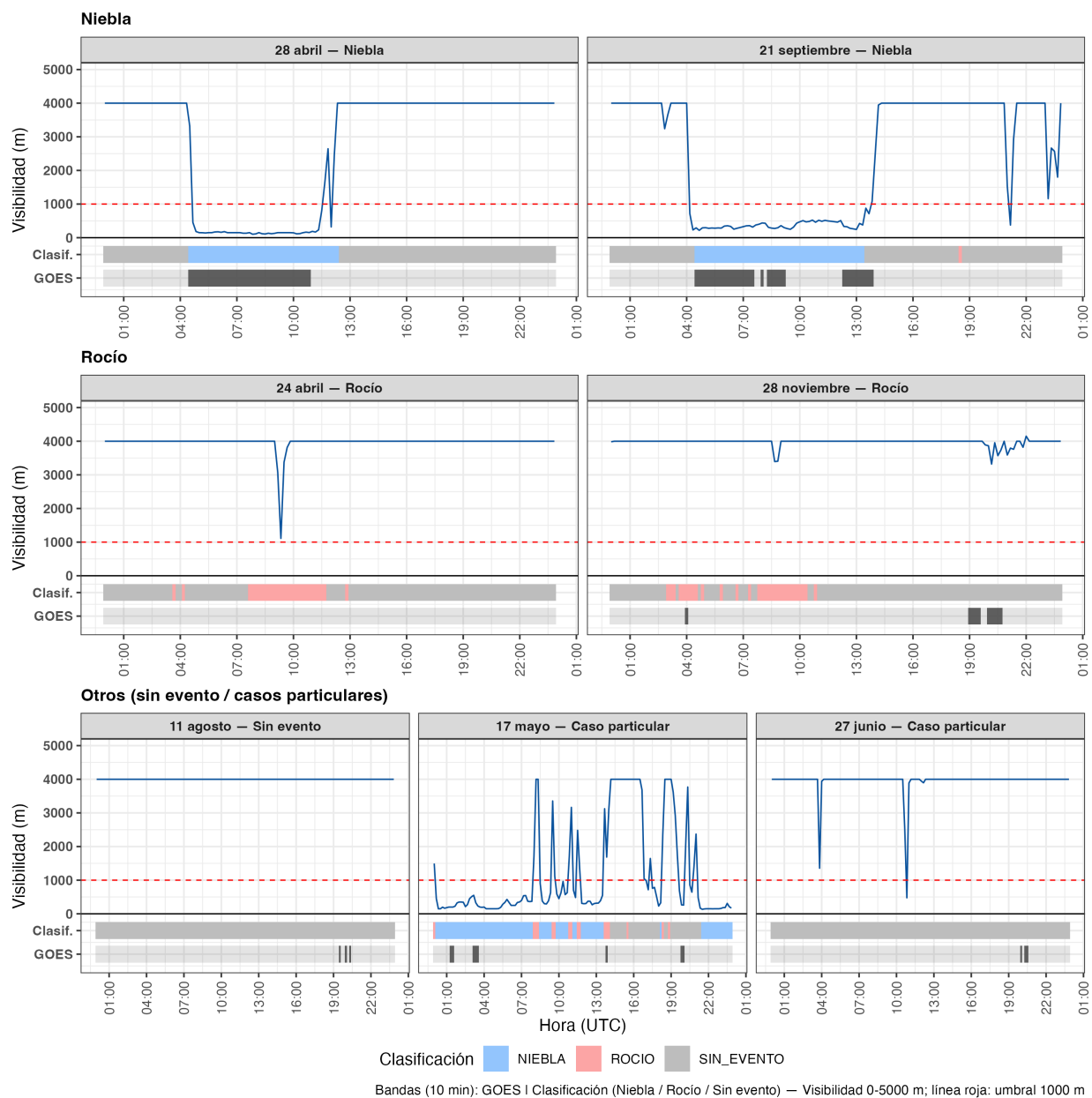
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4 Results and Discussion

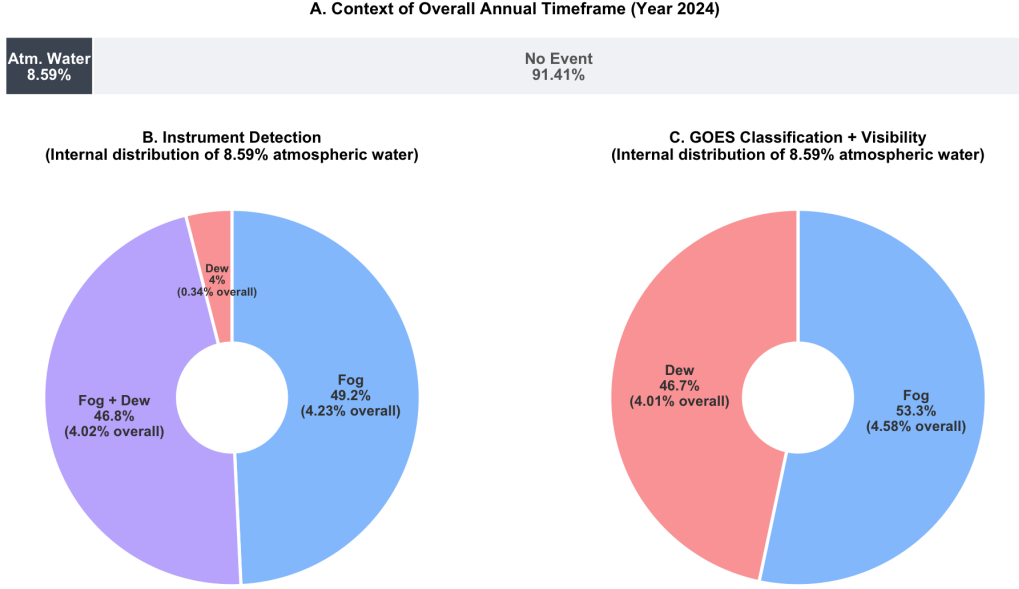
4.1 Reassignment of the atmospheric water event fractions

Comentario: Creo que esta figura anterior debiese sumarle las barritas de los captadores iniciales, solo instrumental, para que se note la diferencia de lo que captó. Esta figura puede ser readaptada igual o quizás si es muy específica no ponerla, pero considero que igual es útil para entender mejor, quizás una figura conjunta con los perfiles verticales de estos mismos días.

- Fig. 4 shows daily examples of the mechanism that motivates this entire section. On two days of unambiguous fog and two of unambiguous dew, visibility, the GOES FLC band, and the final classification clearly agree; on a day without events there is no signal at all; and in one particular case (May 17) erratic visibility alternates between fog and dew within the same day.
- What matters is that, on several of these days, the SDC or the SFC records water on days that the independent classifier assigns to the *opposite* phenomenon — this is the raw instrumental



overlap, visible case by case before aggregating it at the annual scale. *esto anterior es a lo que me refiero que debiese sumar barritas horizontales*

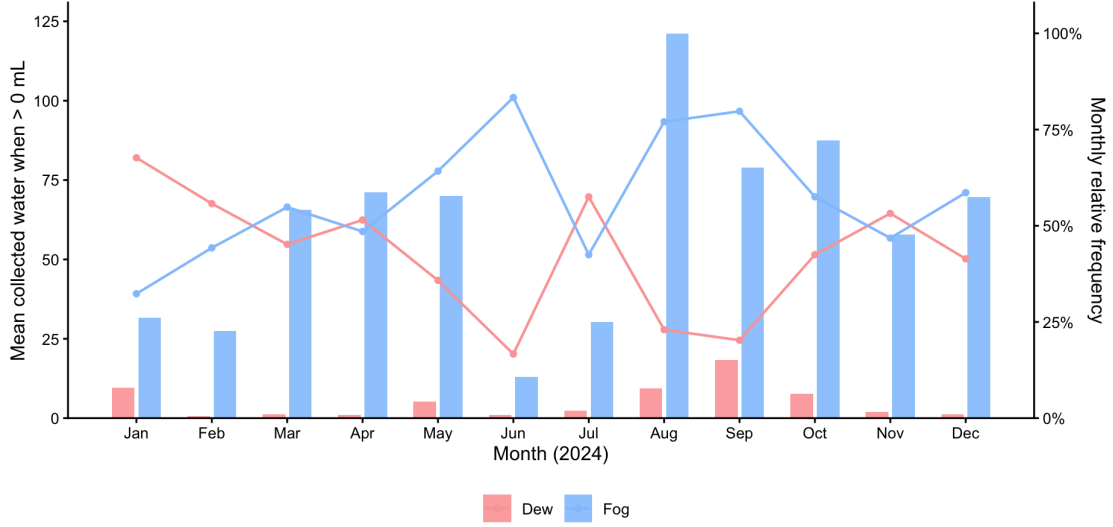


- Fig. 5 (above) aggregates exactly that (as noted in the previous point) across all of 2024. Its top bar first situates the problem at its real magnitude: the active atmospheric-water window occupies only 8.6% of the year (91.4% no-event), so that each misclassified interval carries proportionally more weight than its duration suggests.
- Within that window, direct instrumental detection reports 49.2% fog, 46.8% mixed fog+dew, and just 4.0% pure dew; replacing that criterion with the Visibility+GOES classification (binary by design) shifts the partition to 53.3% fog and 46.7% pure dew.
- In other words, the instrumental overlap is not isolated noise: correcting it multiplies the pure-dew time fraction by more than 10 (4.0% $\rightarrow$ 46.7%). H1 is supported, and this is the central quantitative figure of the paper, from which all the following figures follow.
- The result nonetheless has two limitations worth acknowledging without invalidating it: the visibility sensor itself fogs up during dew (a slight dip that never crosses the fog threshold, visible in the dew panels of Fig. 4), and GOES-16's coarser resolution means a detected low-cloud pixel does not necessarily intersect the topography.

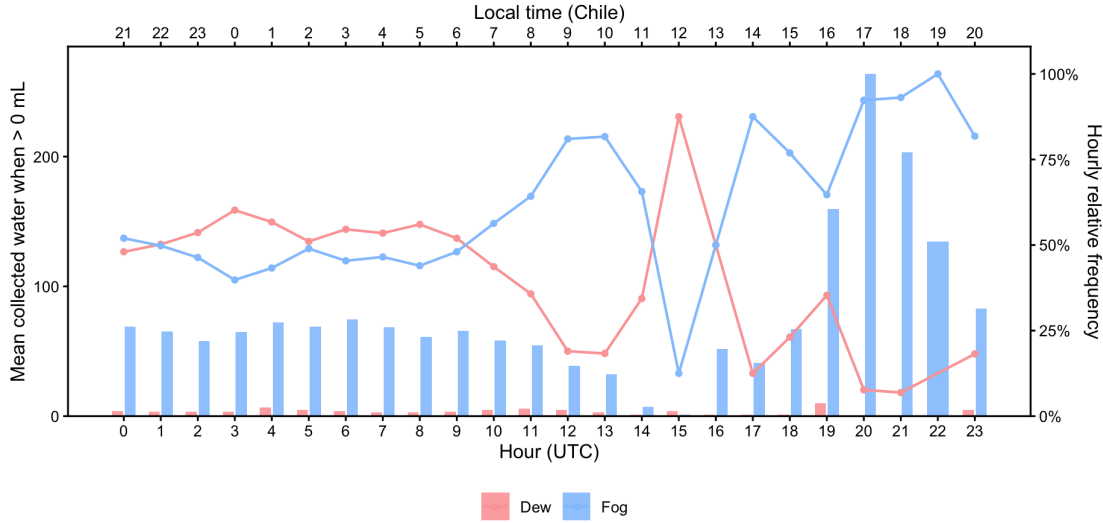
4.2 Seasonal and diurnal distribution of the reclassified events

quizás se podrían pegar estas dos figuras en una?

- This figure shows that average fog collection exceeds dew collection in all twelve months of the year, with August–October concentrating the maximum volumes of both phenomena.



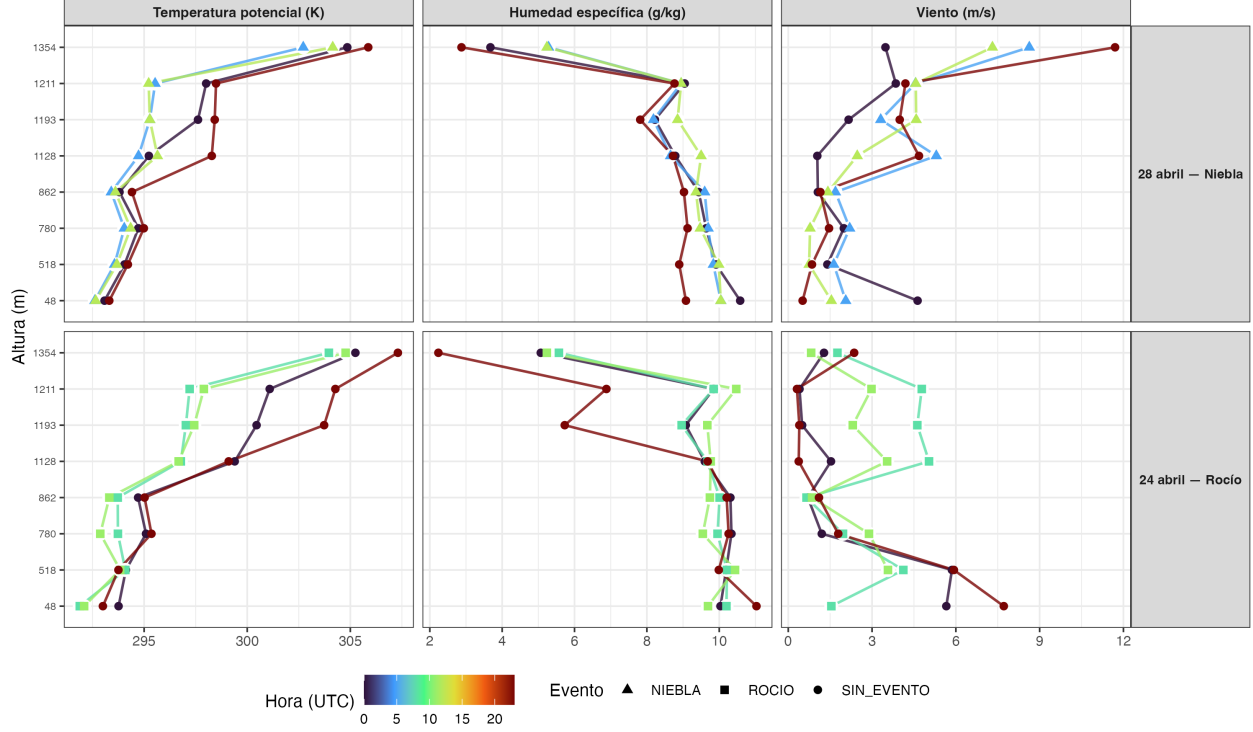
- But magnitude and frequency do not always coincide: in June fog dominates in frequency without yielding much per event, and in August–September dew is comparatively scarce in frequency though occasionally intense — an early signal that counting events and summing the volume they yield are two distinct questions, revisited below.



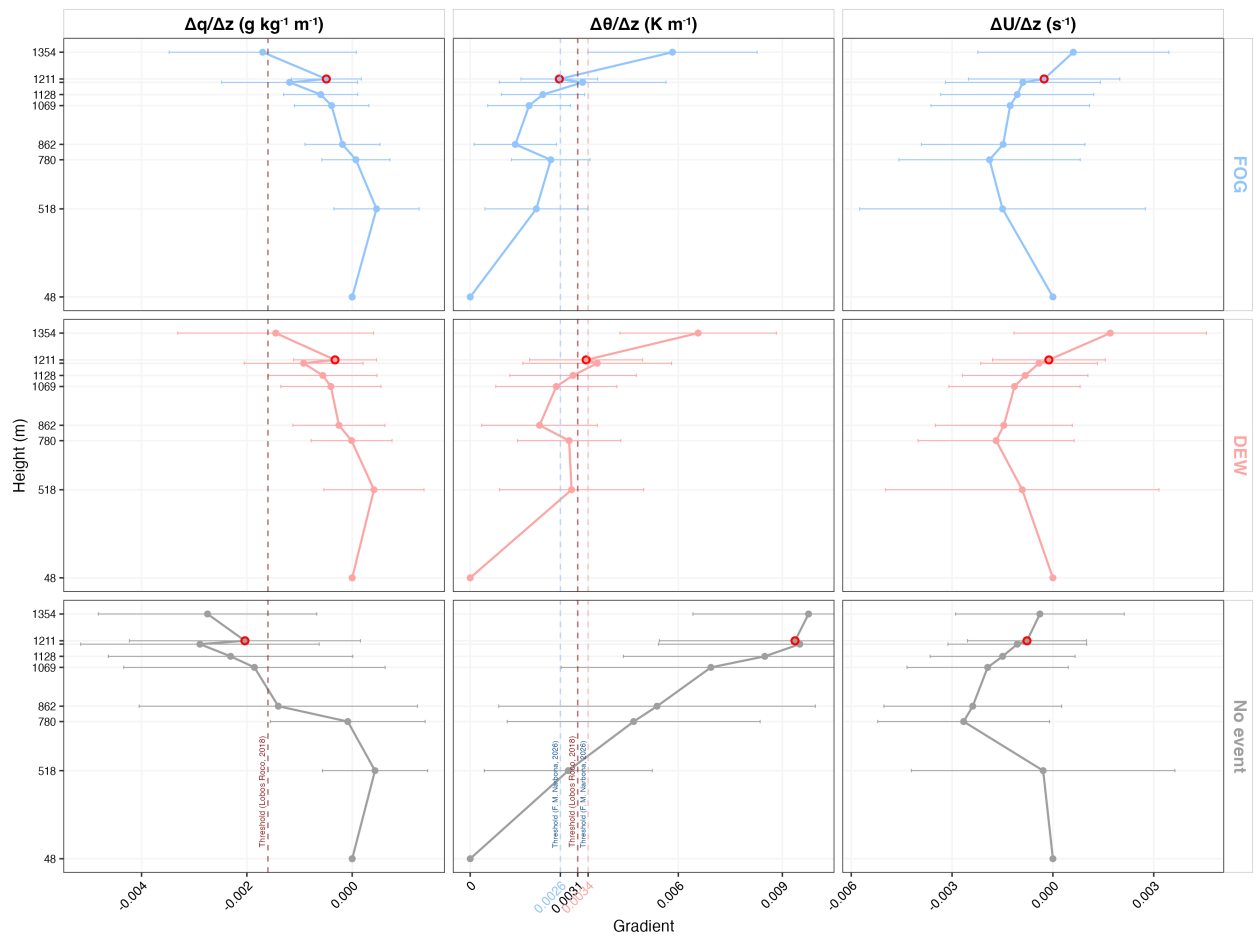
- That same dissociation repeats at the daily scale in the figure: dew sustains its most stable regime between 21:00 and 09:00 local time (00:00–12:00 UTC), with a high and constant relative frequency that exceeds fog during that window.
- Fog, in contrast, dominates between 09:00 and 20:00 local time both in frequency and collected volume, with a peak near 20:00 UTC (~300 mL per 10-minute interval), a day/night asymmetry consistent with the advection versus radiative-cooling mechanisms.
- A punctual spike at 15:00 UTC (12:00 local time) reflects a low sample size in that interval, not a real physical signal.

4.3 Vertical thermodynamic structure of the boundary layer

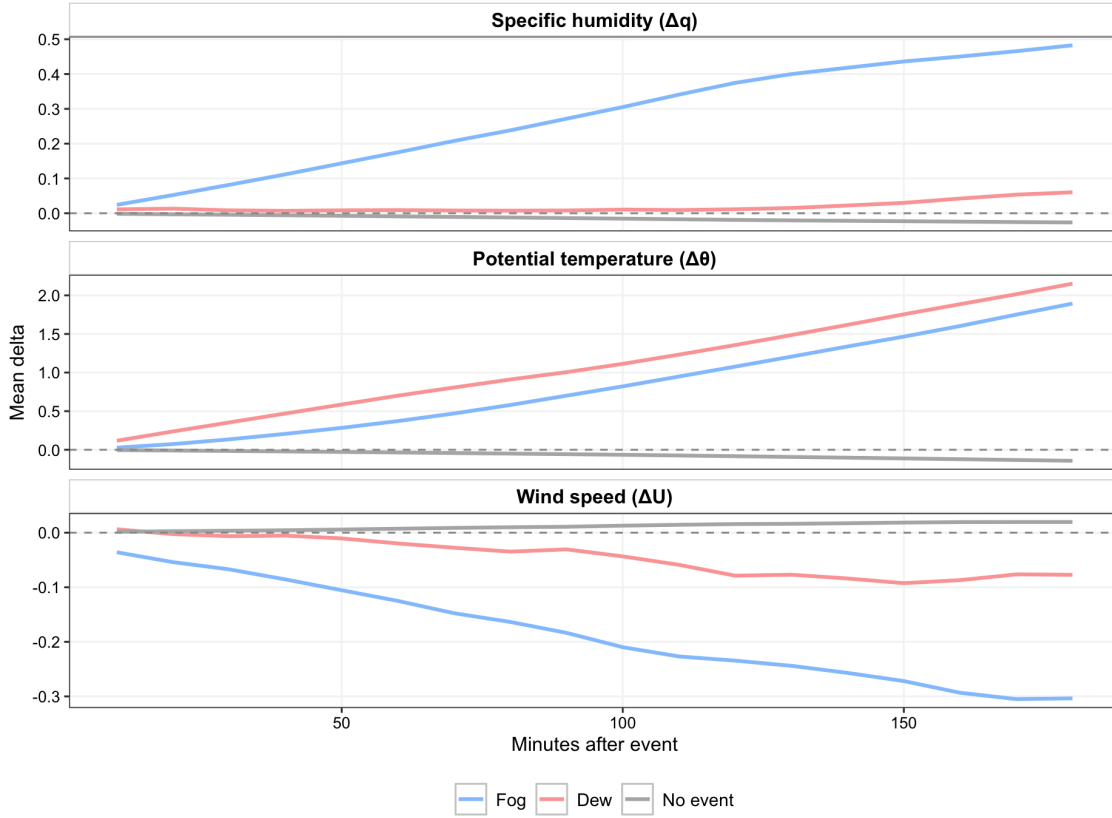
Comentario: esta figura diaria siguiente es la que podría fusionarse con la que pongo dias de ejemplo más arriba, en la clasificación vis+goes



- At the scale of a single day, the figure above already reveals the pattern: April 28 (fog) exhibits a nearly uniform θ profile between 48 and 1354 m a.s.l. (the signature of a well-mixed layer), whereas April 24 (dew) shows a slope similar to that of the no-event hours of that same day. The q and U profiles, in contrast, show no equivalent contrast between the two regimes.
- The figure above generalizes this pattern to the 8 stations along the transect throughout 2024: fog concentrates at $\Delta\theta/\Delta z < 0.0026 \text{ K m}^{-1}$, dew at $0.0026\text{--}0.0034 \text{ K m}^{-1}$, and no-event jumps to $\approx 0.0094 \text{ K m}^{-1}$ — three distinct regimes.
- Mann-Whitney/Kruskal-Wallis tests confirm that this contrast is robust for θ ($r = 0.25$, $p < 0.001$), but not for q or U ($r \approx 0.04\text{--}0.13$) $\rightarrow$ H2 is fully supported, including its negative prediction.
- This repositions the general threshold from Lobos-Roco et al. (2018), 0.0031 K m^{-1} : in this dataset it falls *within* the dew range, rather than at the boundary between dew and no-event as previously assumed. That value separated fog from “everything else” without distinguishing, within that remainder, a proper dew regime.
- The physical reason: both phenomena require a nearly saturated, well-mixed vapor column to initiate, so discrimination cannot rely on moisture — only thermal stability separates the regimes. As a secondary result, an updated $\Delta q/\Delta z$ threshold is also proposed to distinguish active event from no-event ($-0.0004 \text{ g kg}^{-1}\text{m}^{-1}$), refining (without replacing) the primary θ -based criterion.



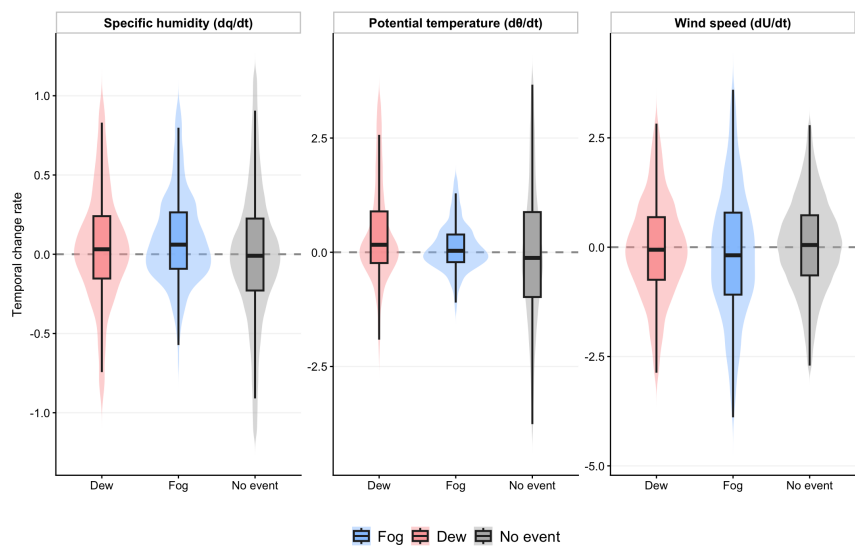
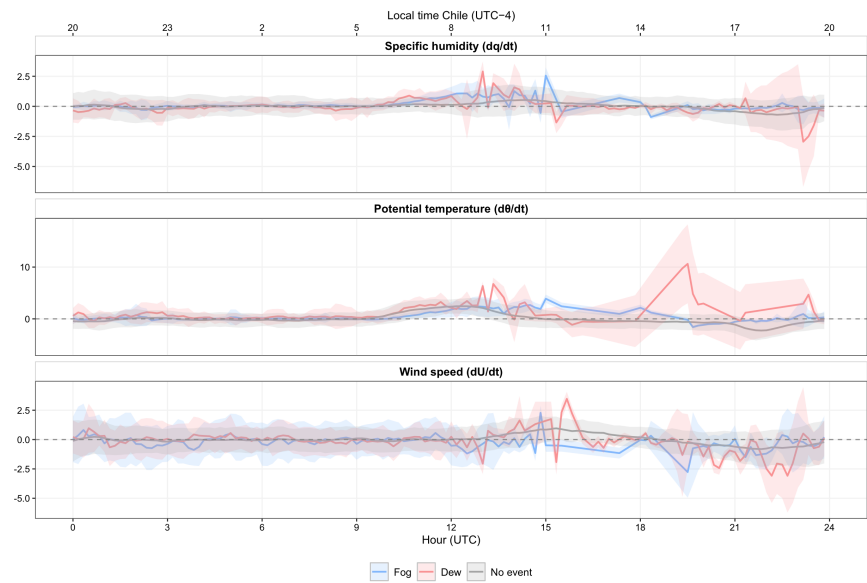
4.4 Temporal evolution and moisture/heat flux signatures



- The figure above answers that question with all three variables at once. Specific humidity behaves oppositely in each regime: fog accumulates $+0.48 \text{ g kg}^{-1}$ by 180 minutes after event onset, a sustained increase consistent with continuous marine moisture advection, while dew barely reaches $+0.06 \text{ g kg}^{-1}$, close to neutral rather than the marked drying that would be expected from an active sink.
- Potential temperature also shows a strong contrast: dew exceeds fog at all times ($+2.15$ versus $+1.89 \text{ K}$ at 180 min), consistent with an episodic, radiatively forced mechanism rather than an advective and steady one.
- Regarding wind: fog clearly decelerates (-0.30 m s^{-1}) as the cloud layer stabilizes the flow, while dew remains nearly neutral (-0.08 m s^{-1}), without reconfiguring the background wind field.
- Taken together, the *sign* of the moisture trend matches what was predicted for both regimes, although the magnitude of the dew sink turns out to be smaller than previously reported for the region: H3 is supported in direction and refined in magnitude, and it is the θ and U signatures (not q alone) that most strongly sustain the advective-vs-radiative distinction.

Comentario: las siguientes figuras podrían fusionarse igual quizás

- The dew moisture sink is weaker than expected and its θ variability is high, pointing to a real but episodic and locally forced mechanism, consistent with the seasonal/diurnal decoupling already shown in the previous figures *** unsure



4.5 Extra

- All these thresholds and rates were derived under the 2024 ENSO context (El Niño → neutral transition), so their interannual generalization to other phases remains an explicit open question, not an implicit assumption.
- And although fog remains the dominant phenomenon by volume (Fig. 6), the ecohydrological relevance of dew rests not on its yield but on its temporal persistence — 46.7% of the active window (Fig. 5) — which is, ultimately, the reason it deserves sustained study despite its modest volumetric contribution.

5 Conclusions

- The hierarchical visibility + GOES-16 classification resolves the collectors' instrumental overlap; the pure-dew time fraction increases from 4.0% (instrumental record) to 46.7% of the active atmospheric-water window → *H1 confirmed*.
- $\Delta\theta/\Delta z$ is the primary physical discriminator between fog and dew regimes, while $\Delta q/\Delta z$ and $\Delta U/\Delta z$ act as shared background conditions rather than discriminators → *H2 confirmed*
- Fog carries an advective, continuous marine moisture signature; dew carries a radiative, thermally episodic signature, with a more moderate moisture sink than previously reported for the region → *H3 confirmed in direction, corrected in magnitude*.
- These results answer the research question: fog and dew do not differ in the boundary layer's moisture or wind conditions, but in its thermal stability and in the advective versus radiative pathway that produces it.
- This reproducible, open-access analytical workflow positions Cerro Oyarbide as a long-term ecohydrological and boundary-layer monitoring platform, ready to incorporate future years of data and, potentially, other ENSO phases.